Introduction to AWS Lambda

SPL-88 Version 2.3.36

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Note: Do not include any personal, identifying, or confidential information into the lab environment. Information entered may be visible to others.

Corrections, feedback, or other questions? Contact us at AWS Training and Certification.

Lab overview

The lab provides a basic explanation of AWS Lambda. It will demonstrate the steps required to get started to create a Lambda function in an event-driven environment.

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources for you, making it easy to build applications that respond quickly to new information. AWS Lambda starts running your code within milliseconds of an event such as an image upload, in-app activity, website click, or output from a connected device. You can also use AWS Lambda to create new back-end services where compute resources are automatically triggered based on custom requests.

Objectives

By the end of this lab you will be able to:

- Create an AWS Lambda function
- Configure an Amazon S3 bucket as a Lambda Event Source
- Trigger a Lambda function by uploading an object to Amazon S3
- Monitor AWS Lambda S3 functions through Amazon CloudWatch Log

Icon key

Various icons are used throughout this lab to call attention to certain aspects of the guide. The following list explains the purpose for each one:

- **Caution:** Important information to prevent issues or unexpected results.
- **Note:** A note, tip, or important guidance.
- **Task complete:** A conclusion or summary point in the lab.
- **Warning:** Critical information to prevent data loss, security issues, or system damage.

Start lab

1. To launch the lab, at the top of the page, choose **Start Lab**.

Caution: You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose **Open Console**.

You are automatically signed in to the AWS Management Console in a new web browser tab.

Task complete: Do not change the Region unless instructed.

Common sign-in errors

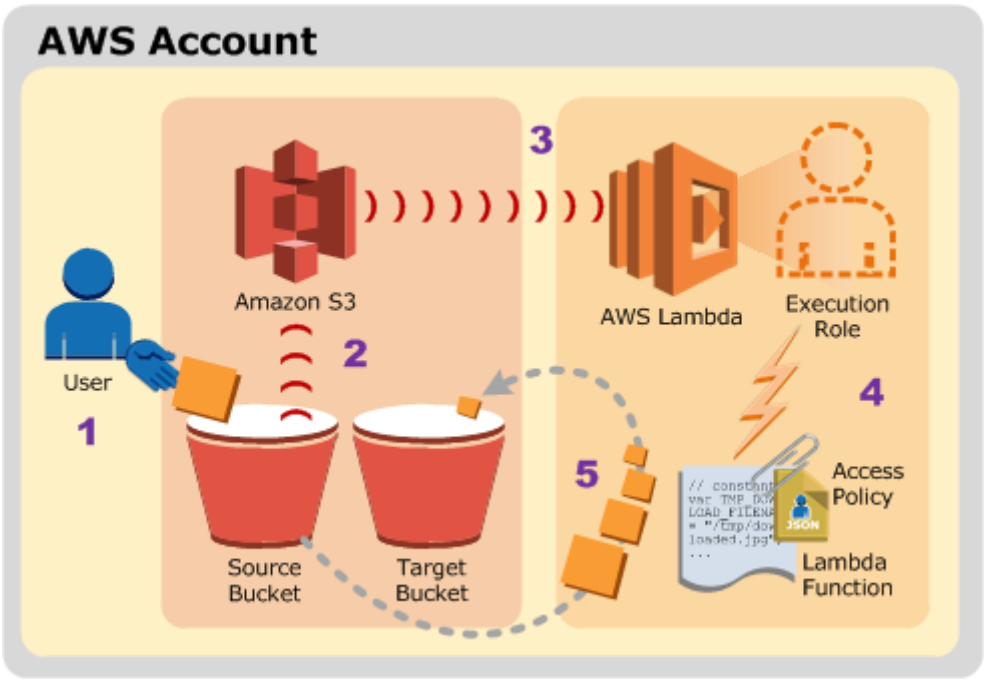
Error: Choosing Start Lab has no effect

- Add the lab domain name to your pop-up or script blocker's allow list or turn it off.
- Refresh the page and try again.

Lab environment

This lab demonstrates AWS Lambda by creating a serverless **image thumbnail application**.

The following diagram illustrates the application flow:

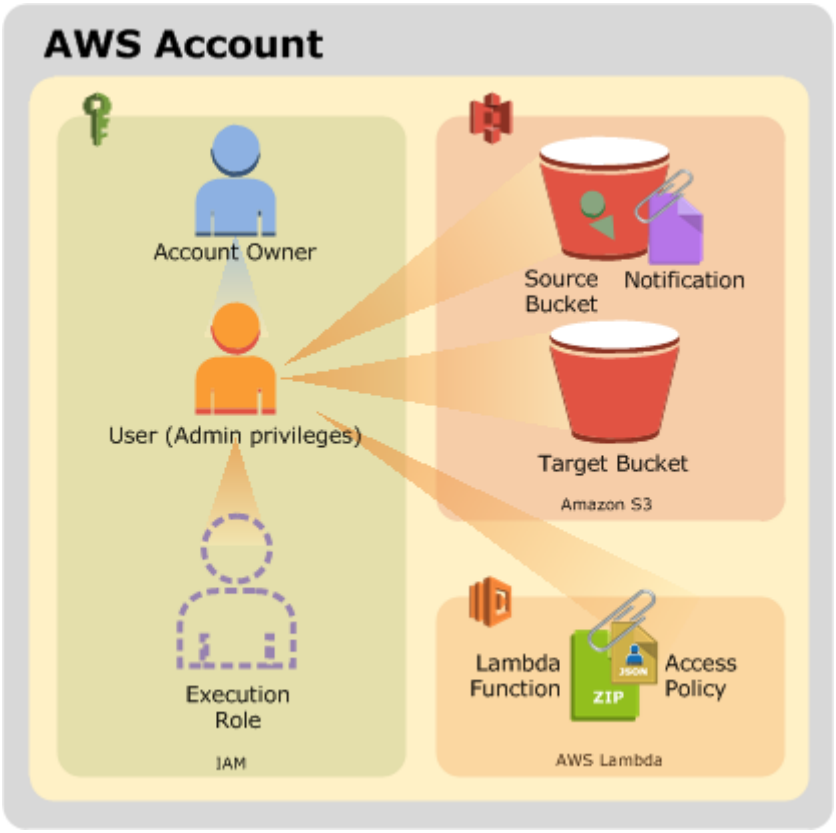


Application flow: an object-created event in Amazon S3 invokes Lambda, which processes the image and stores a thumbnail.

1. A user uploads an object to the source bucket in **Amazon S3** (object-created event).
2. Amazon S3 detects the object-created event.
3. Amazon S3 publishes the object-created event to AWS Lambda by invoking the Lambda function and passing event data as a function parameter.
4. AWS Lambda executes the Lambda function.
5. From the event data it receives, the Lambda function knows the source bucket name and object key name. The Lambda function reads the object and creates a thumbnail using graphics libraries, then saves the thumbnail to the target bucket.

Upon completing this tutorial, you will have the following resources in your account:

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Lab resource overview: the user, S3 buckets, Lambda function, execution role, and access policy used by the lab.

The steps in this lab show you how to create the Amazon S3 buckets and the Lambda function. You then test the service by uploading images for resizing.

Task 1: Create the Amazon S3 buckets

In this task, you create two Amazon S3 buckets – one for input and one for output.

3. At the top of the AWS Management Console, in the unified search bar, search for and choose **S3**.
4. Choose **Create bucket**.
5. For **Bucket name:**, enter `images-randomnumber`, but replace **randomnumber** with an actual random number to ensure a unique bucket name such as *images-123456789*.

> **Note:** Amazon S3 buckets require unique names regardless of region.

6. Copy the name of your bucket to a text editor for later use.
7. Keep everything else as default and choose **Create bucket** (at the bottom of the screen).

> **Note:** If you receive an error stating The requested bucket name is not available, then choose the first Edit link, change the bucket name to a different random number and try again until it works.

8. Create another bucket for the output. Choose **Create bucket** and create another bucket with the same name as the first, but with `-resized` appended to the name of this new bucket, such as *images-123456789-resized*.

You should now have buckets named something similar to **images-123456789** and **images-123456789-resized**.

9. Download the file **HappyFace.jpg** to your local machine (*Save link as…*).
   - HappyFace.jpg
10. Open the image on your computer.

It is a large picture, with dimensions of 1280 × 853.

11. In the AWS console, choose the link to the first bucket you created (the one without the *-resized* appendix).
12. Choose **Upload**.
13. Choose **Add files**.
14. Use your file explorer and select the **HappyFace.jpg** file you just downloaded to your local machine.
15. Choose **Upload**.

> **Note:** Later in this lab you invoke a Lambda function manually by passing sample event data to the function. The sample data refers to this HappyFace.jpg image.

> **Task complete:** You have created an S3 bucket and stored an object on it.

Task 2: Create an AWS Lambda function

In this task, you create an AWS Lambda function that reads an image from Amazon S3, resizes the image, and then stores the new image in Amazon S3.

16. At the top of the AWS Management Console, in the unified search bar, search for and choose **Lambda**.
17. Choose **Create function**.
18. On the **Create function** page, select **Author from scratch**.

> **Note:** Blueprints are code templates for writing Lambda functions. Blueprints are provided for standard Lambda triggers such as creating Alexa skills and processing Amazon Kinesis Firehose streams. This lab provides you with a pre-written Lambda function, so you Author from scratch.

19. In the **Basic information** section, for **Function name:**, enter `Create-Thumbnail`.
20. For **Runtime:**, select `Python 3.12`.
21. Expand the **Additional settings** section.
22. In the **General** section, turn on the **Custom execution role** toggle.
23. In the **Configure custom execution role** panel that appears, for **Existing role:**, select `lambda-execution-role`.

> **Note:** This role grants permission to the Lambda function to access Amazon S3 to read and write the images.

24. Choose **Save**.

Note: To make this lab more secure, you must attach your function to a VPC, subnet, and security group that have been created for you. Conventionally, Lambda functions do not need to be attached to a VPC, but this lab will not let you create the function without the addition of defined networking.

25. In the **Additional settings** section, scroll down to the **Networking** section and turn on the **VPC** toggle.
26. In the **Configure VPC** panel, for **VPC**, select the VPC with CIDR range `10.0.0.0/16`.
27. For **Subnets**, select the subnet with CIDR range `10.0.1.0/24`.
28. For **Security groups**, select the security group with **LambdaSecurityGroup** in its name.
29. Choose **Save**.
30. Choose **Create function**.
31. On the **Getting started** pop-up, choose **Dismiss**.

> **Note:** Creating a VPC-attached Lambda function takes a few minutes. Wait until a banner at the top of the page displays Successfully created the function Create-Thumbnail. to proceed.

32. Choose **Add trigger**.
33. For **Trigger configuration**, use the drop-down selector to select **S3**.
34. For **Bucket:**, select your *images-* bucket (e.g. *images-123456789*).
35. For **Event type:**, select **All object create events**.
36. For **Recursive invocation**, select the checkbox for **I acknowledge that …**.
37. Choose **Add**.
38. Choose the **Code** tab.
39. Choose **Update** and select **Update from a file in Amazon S3**.
40. In the **Update from a file in Amazon S3** dialog box, for **Code storage mode**, select **Copy mode (default)**.
41. For **Amazon S3 link**, copy and paste the value for `AmazonS3LinkURL` from the panel to the left of these instructions.
42. Choose **Update**.

The function package contains code that receives an event, downloads the image to local storage, resizes it using the Pillow library, and uploads the resized image to the *-resized* bucket.

Lambda function code

```
import boto3
import os
import sys
import uuid
from PIL import Image
import PIL.Image

s3_client = boto3.client('s3')

def resize_image(image_path, resized_path):
    with Image.open(image_path) as image:
        image.thumbnail((128, 128))
        image.save(resized_path)

def handler(event, context):
    for record in event['Records']:
        bucket = record['s3']['bucket']['name']
        key = record['s3']['object']['key']
        download_path = '/tmp/{}{}'.format(uuid.uuid4(), key)
        upload_path = '/tmp/resized-{}'.format(key)

        s3_client.download_file(bucket, key, download_path)
        resize_image(download_path, upload_path)
        s3_client.upload_file(upload_path, '{}-resized'.format(bucket), key)
```

43. In the **Runtime settings** section, choose **Edit**.
44. For **Handler**, replace the existing text with `CreateThumbnail.handler`.
45. Choose **Save**.

> **Note:** Make sure you set the Handler field to CreateThumbnail.handler, otherwise the Lambda function will not be found.

46. Choose the **Configuration** tab.
47. Choose **General configuration** on the left, then choose **Edit**.
48. For **Description - optional**, enter `Create a thumbnail-sized image`.
49. Choose **Save**.

> **Note:** Leave the other settings as default. Memory defines the resources allocated to the function; increasing memory also increases CPU allocated to the function. Timeout sets the maximum duration for function execution.

> **Task complete:** Your Lambda function has now been configured.

Task 3: Test your function

In this task, you test your Lambda function. This is done by simulating an event with the same information normally sent from Amazon S3 when a new object is uploaded.

50. Choose the **Test** tab.
51. In the **Test** event section, select **Create new event** and then configure:
52. For **Event name:**, enter `Upload`.

53. For **Template - optional:**, select `S3 Put`.
54. Replace `example-bucket` text in the **Event JSON** section with the name of your images bucket (e.g. *images-123456789*) that you copied to your text editor.
55. Replace `test%2Fkey` text in the **Event JSON** section with the name of the picture that you uploaded. This should be **HappyFace.jpg**.
56. Choose **Test**.

Note: The example-bucket text appears twice, as the bucket name and the bucket ARN. Replace only the example-bucket text and leave the rest of the ARN intact. Be sure to replace it in both locations.

Note: The test%2Fkey value is the object key in the JSON, near the bottom.

After a moment, you should see the message: *Executing function: succeeded.*

Choose **Details** to expand it. You are shown information including execution time, resources configured, maximum memory used, and log output.

57. At the top of the AWS Management Console, in the unified search bar, search for and choose **S3**.
58. Choose the name of your *-resized* bucket (which is the second bucket you created).
59. Select the checkbox for **HappyFace.jpg**.
60. Choose **Open**.

Note: Depending on your browser and settings, opening the image may either display it or download it. If it is downloaded, open it on your machine using the appropriate app.

The image should now be a smaller thumbnail of the original image. You are welcome to upload your own images to the *images-* bucket and then check for thumbnails in the *-resized* bucket.

Task complete: You have triggered a Lambda function and observed the results.

Task 4: Monitoring and logging

In this task, you monitor AWS Lambda functions to identify problems and view log files to assist in debugging.

61. At the top of the AWS Management Console, in the unified search bar, search for and choose **Lambda**.
62. Choose your **Create-Thumbnail** function (the link, not the checkbox).
63. Choose the **Monitor** tab.

The console displays graphs including:

- **Invocations:** The number of times that the function was invoked.
- **Duration:** The average, minimum, and maximum execution times.
- **Error count and success rate (%):** The number of errors and the percentage of executions that completed without error.

- **Throttles:** When too many functions are invoked simultaneously, they will be throttled. The default is 1000 concurrent executions.
- **Async delivery failures:** The number of errors that occurred when Lambda attempted to write to a destination or dead-letter queue.
- **Iterator Age:** Measures the age of the last record processed from streaming triggers (Amazon Kinesis and Amazon DynamoDB Streams).
- **Total concurrent executions:** The number of function instances that are processing events.

64. Choose **View CloudWatch logs**.
65. Choose the **Log Stream** that appears (the link, not the checkbox).
66. Expand each message to view the log message details.

Log messages from Lambda functions are retained in **Amazon CloudWatch Logs**.

> **Note:** Event data includes the Request Id, duration (in milliseconds), billed duration, Memory Size, Maximum Memory Used, and any logging messages or print statements. This assists in debugging.

> **Task complete:** You have monitored AWS Lambda to S3 functions using Amazon CloudWatch.

Conclusion

- Created an AWS Lambda function
- Configured an Amazon S3 bucket as a Lambda Event Source
- Triggered a Lambda function by uploading an object to Amazon S3
- Monitored AWS Lambda S3 functions through Amazon CloudWatch Log

End lab

67. Return to the AWS Management Console.
68. At the upper-right corner of the page, choose **AWSLabsUser**, and then choose **Sign out**.
69. Choose **End Lab** and then confirm that you want to end the lab.